

Experiment	<i>Lab 05: Experiment SWT – Schlieren Imaging & Supersonic Wind Tunnels</i>
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Section Number	04 – Robert Bryan

Summary

Schlieren imaging was used in this experiment to examine how shock waves develop in a nozzle in the Embry-Riddle supersonic wind tunnel. This experiment's goal was to learn how to use Schlieren photography and data collection to quantify and evaluate flow parameters through a nozzle. Schlieren images taken during the investigation showed that a shock wave was present along the nozzle between positions 9.5 and 10.5 inches. There was an obvious increase in pressure as a result of the shock, suggesting compressive effects common to shock behavior. With a computed shock strength of 1.6142 and a nozzle pressure ratio (NPR) of 3.932, the shock was classified as moderately strong. Furthermore, it was discovered that the mass flow rate via the nozzle was 6.1036 lbm/s.

Aim & Objectives

The aim of this lab was to understand the components of a supersonic wind tunnel.

The objectives are:

- Learn about a supersonic wind tunnel's features and how it operates.
- Calculate the distribution of static pressure along the nozzle's floor.
- Determine the NPR (nozzle pressure ratio).
- Determine the nozzle flow using the experimental data.

Uncertainty Analysis

A 95% confidence interval is used to calculate the uncertainty in all measured values and visualizations.

Results & Discussion

Wind Tunnel Components:

We have to understand the parts of the supersonic wind tunnel and how they work before moving on to the analysis's computational analysis section.



Figure 1.1: Wind Tunnel Dryer

In addition to preventing condensation in the tank, the wind tunnel dryer makes sure that the incoming air supply is completely dry.

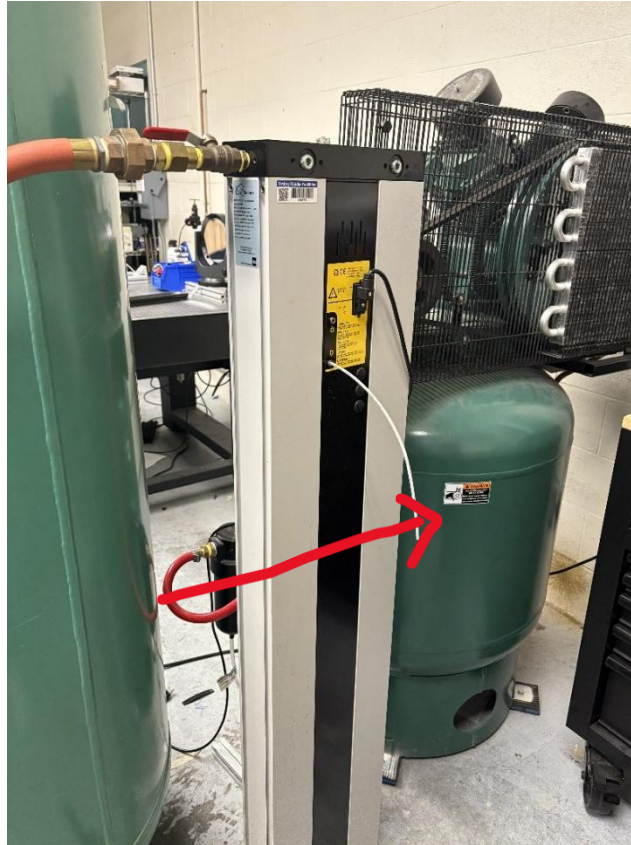


Figure 1.2: Air Compressor

The air is pressurized by the air compressor prior to its passage through the wind tunnel. This is what causes the tunnel's high-speed flow.

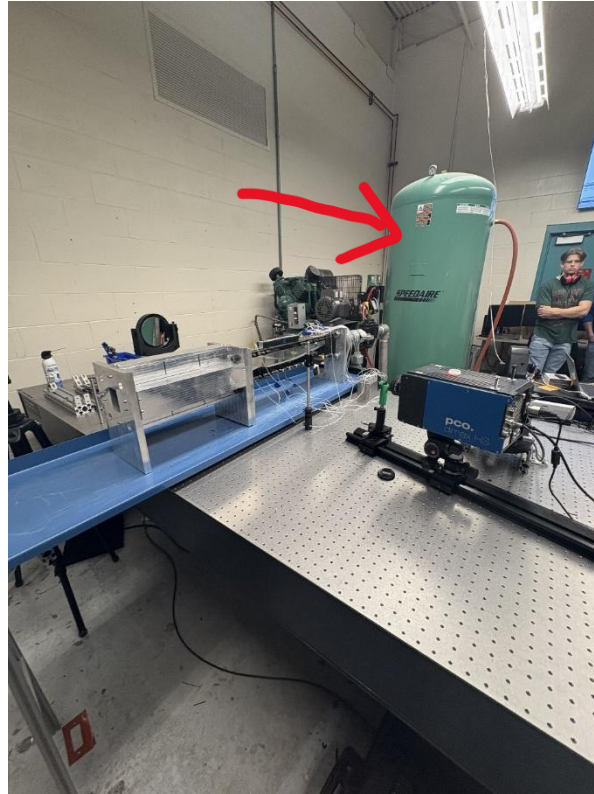


Figure 1.3: Air Storage Tank

Before the pressured air from the compressor is discharged into the wind tunnel, it is first stored in the air storage tank.



Figure 1.4: Main Valve

The compressed air is released from the storage tank under the control of the main valve.



Figure 1.5: Pressure Regulator

To prevent overflow into the wind tunnel, the regulator valve keeps the pressure steady.

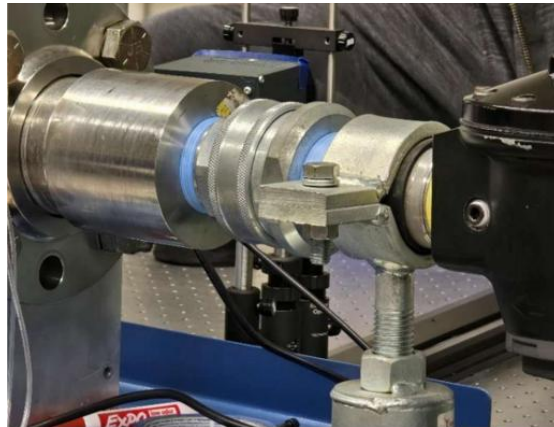


Figure 1.6: Wind Tunnel Quick-Disconnect

To perform a variety of studies, the quick-disconnect enables the rapid interchangeability of different nozzle types.

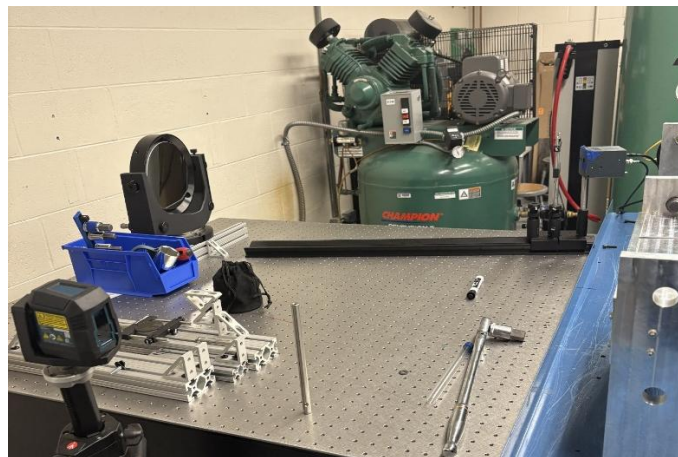


Figure 1.7: Schlieren Imaging System

An optical technique called Schlieren imaging is used to see variations in air density in high-speed flows, such as expansion fans and shock waves. It detects how light rays bend as they travel through areas with different refractive indices, producing patterns of light and dark in the finished image. This makes it possible to observe flow characteristics in a supersonic wind tunnel without causing any disturbance.

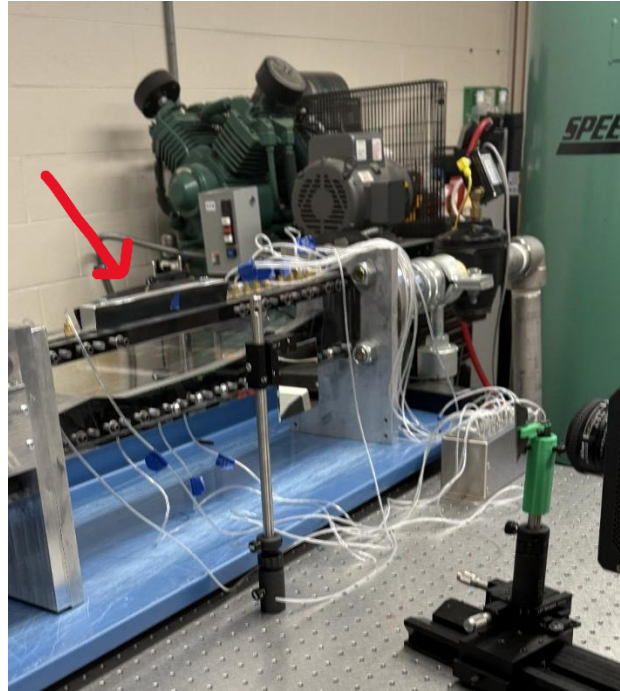


Figure 1.8: Supersonic Wind Tunnel Nozzle

This nozzle speeds up the airflow to supersonic levels using convergent and divergent sections. Pressure taps are used along the test section's length to measure the separation.

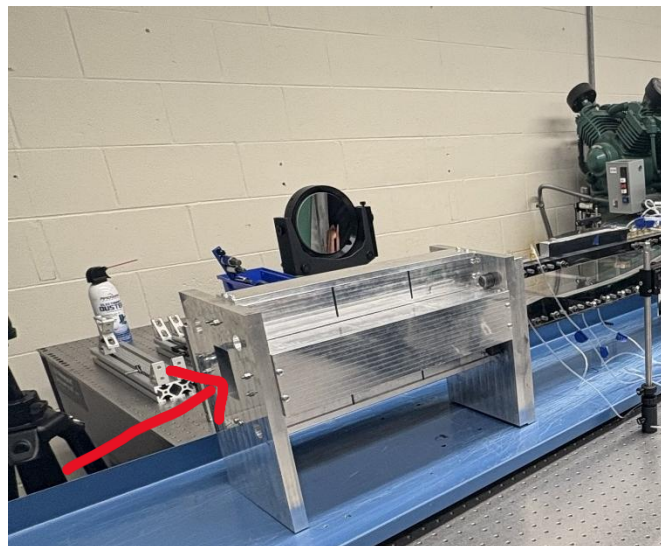


Figure 1.9: Wind Tunnel Silencer

The noise generated by the high-speed flow leaving the wind tunnel gets reduced by the silencer.

Schlieren-Imaging Analysis:

We can see the shock inside the nozzle by using the Schlieren-imaging technology. Illustrations of the shock produced in the nozzle can be found below:

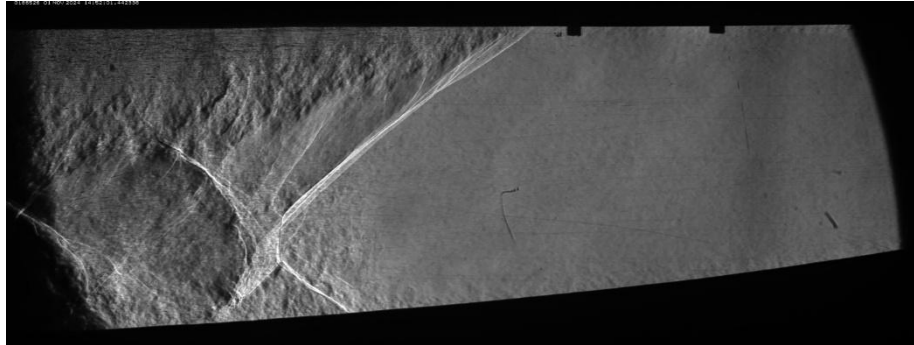


Figure 2.1: Schlieren Image of Shock

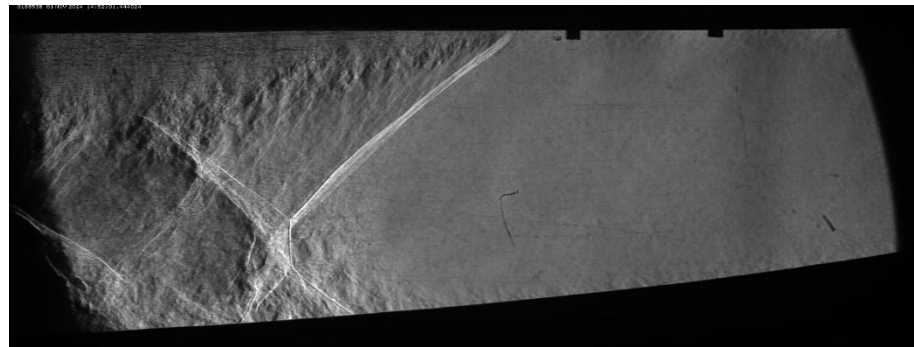


Figure 2.2: Schlieren Image of Shock

Nozzle Pressure Ratio:

The following formula is used to get the nozzle pressure ratio:

$$NPR = \frac{P_0}{P_{exit}}$$

This experiment's NPR was estimated to be roughly 3.932. We can ascertain the flow regime and shock formation by computing the NPR. The flow has been shocked since the NPR is higher than the crucial value.

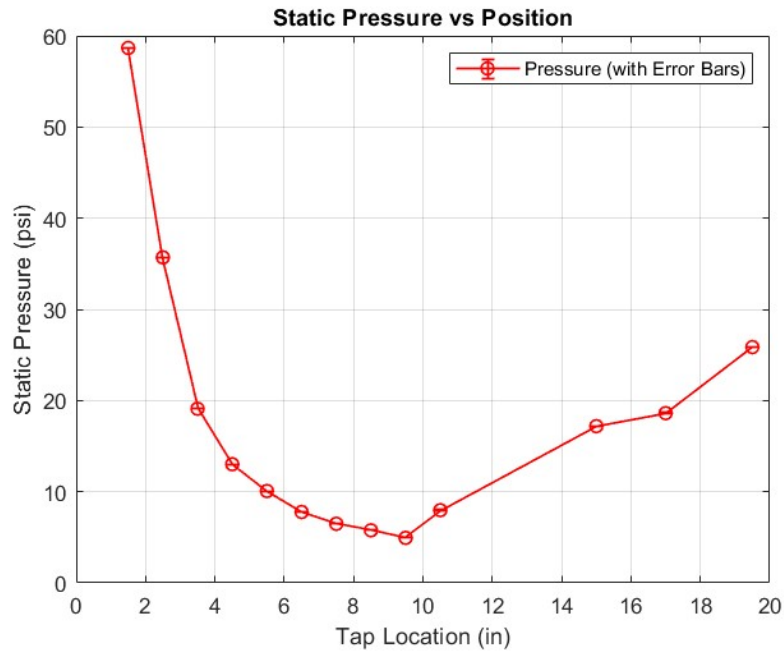
Static Pressure Distribution:

Figure 3.1: Pressure vs Position in Wind Tunnel

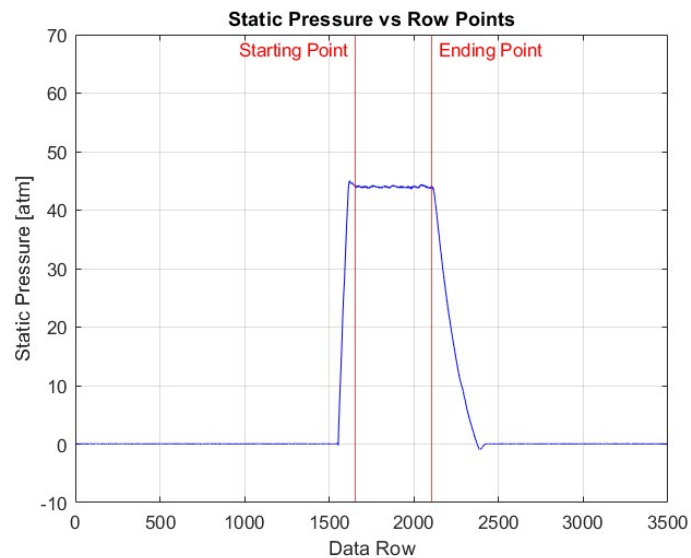


Figure 3.2: Pressure Distribution vs Row Points

Shock Strength:

This formula is used to determine shock strength:

$$S = \frac{P_{\text{before separation}}}{P_{\text{after separation}}}$$

The shock strength in this experiment was determined to be 1.6142. This indicates that the shock is moderate.

Mass Flow:

The following formula is used to determine the mass flow via the nozzle:

$$\dot{m} = A_t \frac{P_0}{\sqrt{T_0}} \sqrt{\frac{\gamma}{R}} \left(\frac{\gamma + 1}{2} \right)^{\frac{\gamma+1}{2(\gamma-1)}}$$

It was determined that the mass flow for this experiment was 6.1036 pounds per second. The examination of the supersonic wind tunnel lab is now complete.

References

1. Experiment SWT – Schlieren Imaging & Supersonic Wind Tunnels: (2025). Embry-Riddle Aeronautical University.
2. AE315 Supplement – Experimental Aerodynamics Supplement: (2025). Embry-Riddle Aeronautical University.

Appendix

Appendix C: MATLAB Code

```
% Author: Anuranan Bharadwaj
% Lab 05: Experiment SWT

clc;
clear;
close all;

%% Data Reading
data = readtable('Pressure_data.csv');
tempData = table2array(data(:, 3:15));

% Constants
Patm = 14.7;           % psi
T0 = 27;               % °C
T = T0*(9/5) + 491.67; % Rankine
gamma = 1.4;
R = 1716;
Ath = 0.765;          % Throat area (in^2)
lower_end = 1653;
upper_end = 2103;

% Building Dataframe
range_steadyState = tempData(lower_end:upper_end, :);
dataframe = mean(range_steadyState) + Patm;

%% Total Pressure

pTaps = [1.5, 2.5, 3.5, 4.5, 5.5, 6.5, 7.5, 8.5, 9.5, 10.5, 15, 17, 19.5];
shock = dataframe(10) / dataframe(9);
pTotal = dataframe(2);
Mach = 1;
p_P0 = (1 + (gamma-1)/2 * Mach^2)^(-(gamma/(gamma-1)));
P0_p = 1 / p_P0;
P0 = P0_p * pTotal;

%% Nozzle

Pb = mean(dataframe(11));
NPR = P0 / Pb;
a = sqrt(gamma * R * T);
Ue = a * sqrt((2 / (gamma-1)) * (1 - (NPR^((1-gamma)/gamma))));

%% Mass Flow
g = 32.174;
mdot = Ath * gamma * (P0 / a) * ((gamma+1)/2)^((gamma+1)/(gamma-1)) * g;
sinTheta = sind(4.8);
pDiff = dataframe(10:end) - dataframe(11);
integral = pDiff * sinTheta;
Cf = 1 + 1/(mdot*Ue) * trapz(pTaps(10:end), integral);
```

```
%% Plotting
% Pressure vs Data Taps
figure(1)
plot(1:size(tempData(:,1),1), tempData(:,1), 'b-')
hold on
xline(lower_end, 'r', 'Starting Point', 'LabelOrientation', 'horizontal', 'LabelHorizontalAlignment', 'left')
xline(upper_end, 'r', 'Ending Point', 'LabelOrientation', 'horizontal', 'LabelHorizontalAlignment', 'right')
xlim([0, 3500])
ylim([-10, 70])
xlabel('Data Row')
ylabel('Static Pressure [atm]')
title('Static Pressure vs Row Points')
grid on
hold off

% Pressure Distribution
uncertainty = 1.96 * std(range_steadyState) ./ sqrt(size(range_steadyState, 1));
figure(2)
errorbar(pTaps, dataframe, uncertainty, 'or-', 'LineWidth', 1)
xlabel('Tap Location (in)')
ylabel('Static Pressure (psi)')
title('Static Pressure vs Position')
legend('Pressure (with Error Bars)')
grid on
```